Inhwa Song • Statement of Purpose

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One in two people globally will experience a mental health disorder over the course of their lifetime. My interest in mental health began in elementary school when I received light peer counseling training. Over the years, I continued my training as a peer counselor, which allowed me to witness firsthand how deeply mental health experiences are tied to an individual's unique socioeconomic and cultural background. Additionally, serving as a student president during the COVID crisis, I heard the voices of international students who were marginalized by non-inclusive health policies and technologies within Korea's homogeneous cultural context. These experiences collectively shaped my vision toward inclusive mental health infrastructure, such as technology design and policies. I am excited to examine this aim at the intersection of human-computer interaction (HCI), social computing, and computational social science (CSS) through my doctoral studies. To this end, I am eager to pursue my Ph.D. at Georgia Tech Human-Centered Computing and receive the tutelage of Prof. Munmun De Choudhury.

My research interest has both guided and been shaped by my involvement in many exciting research projects across diverse academic domains and industry settings. During my exchange period at Georgia Tech, under the mentorship of **Prof. Munmun De Choudhury**, I led timely research to examine how individuals understand and use LLM chatbots when seeking support for their mental health needs in real-world settings [2]. I conducted semi-structured interviews with 21 participants from every inhabited continent by connecting with local forums and support groups. Grounding analysis in psychotherapy literature of what makes support effective, we introduced the concept of therapeutic alignment. Our work has been successfully submitted to *CSCW'24* with me as the first author, currently undergoing review following a request for revisions. I realized that providing interaction components for the user to gain control within a reliable structure is the key design consideration of building responsible generative AI systems, further motivating me to not only envision but to build and test the reliability of such systems.

This motivation led me to a research internship at NAVER AI Lab to work with **Dr. Young-Ho Kim**, whom I have long admired for his novel system contributions to HCI research. I developed ExploreSelf, an LLM-driven interactive system that supports users in exploring their narratives of personal challenges by actively engaging with adaptive guidance [1], grounding in a formative study. Even with my vast hands-on experience managing the development and deployment of complex computing systems, this particular work was especially valuable to me for three main reasons. First, I gained a deeper understanding of what system contribution means in HCI research. Second, this research I led received a *Revise & Resubmit decision in CHI'25* with compliments for the high research quality. Last but not least, I recognized my ability to quickly learn and apply new technologies and programming methods, along with the confidence that I can effectively tackle any new technical challenges I face in future research.

During my internship at NAVER AI Lab, I could communicate with AI researchers from diverse backgrounds, each bringing a different worldview. For instance, within the goal of responsible AI, stakeholders' perspectives diverged between creating safety benchmarks, embedding safety in LLMs, and investigating what responsible AI means in the context of Korean society. Through direct conversations with stakeholders, I saw firsthand how such divergent views stemmed from the tension between qualitative societal contexts and the quantitative methods that dominate ML. These discussions reinforced my commitment to bridge the communication gap between diverse worldviews. They also motivated me to broaden my own perspective on navigating the balance between technical feasibility and social requirements. While I have experienced translating qualitative insights into concrete system design [2], I also wanted to tackle research questions that involve direct intervention in language models. Under

the guidance of **Prof. Alice Oh**, my ongoing project involves constructing a dialogue dataset of unstructured learner inputs paired with concise queries by fine-tuning the Phi-3 model with the StackExchange dataset, where each post combines a body of long-form explanations and a title of a focused question.

These research projects gave me the opportunity to adopt a wide range of approaches, from system building and evaluation [1,8], large-scale data analysis [4], fine-tuning language models, to qualitative methods such as semi-structured interviews [2] and cultural probe [3], and also systematic reviews [7]. Above all, I have developed my ability to identify valuable research questions and learned how to transform an initial idea into systematic research.

I aspire to continue my vision towards inclusive health technologies at the intersection of HCI, social computing, and CSS. With my potential directions to keep in mind, Human-Centered Computing at Georgia Tech would be an ideal place to pursue my doctoral study. I hope to work with **Prof. Munmun De Choudhury**, who has continuously provided me with insights into the current state and prospective research trajectories. A project she is leading, focused on pediatric mental health interventions using a "whole child" approach, strongly resonates with my research interests. I am particularly inspired by the effort to integrate diverse data sources, from electronic health records to untapped resources such as sensor-based activity tracking and interactions with online technologies. Working under Prof. Munmun De Choudhury's mentorship, I hope to advance the mission to develop predictive models and tools that inform clinical interventions for diverse pediatric populations, directly contributing to improving care outcomes for young patients and their families. This project also perfectly aligns with the workshop that I am organizing, inviting diverse stakeholders to understand the changing needs of the sociocultural and technical landscape of mobile technology and teen mental health [6]. Upon completing my Ph.D., I aspire to remain in academia as a professor, continuing to research and mentor students to advance the sociotechnical landscape of mental health research and real-world impact. I am confident in my ability to succeed in the Georgia Tech HCC Ph.D. program and contribute meaningfully to its mission of inclusive innovation.

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